

9(a)(i)	Simple harmonic motion is defined as an oscillatory motion in which the acceleration of an object is directly proportional to the displacement of the object from its equilibrium position, and the acceleration is always directed towards that position.
9(a)(ii)1.	The graph's positive and negative displacement shows that the mass is oscillating. When it's displacement is positive, its acceleration is negative, and vice versa.
9(a)(ii)2.	The line is not straight showing that displacement is not proportional to acceleration.
9(b)(i)1.	The plate is at its upwards maximum displacement.

9(b)(i)2.	At the maximum upward displacement from free body diagram of sand $mg - N = m\omega x_0^2$ $N = 0$ (when sand just loses contact with surface) $mg = m\omega^2 x_0$ $9.81 = (2\pi f)^2 (x_0)$ $9.81 = (2\pi(13))^2 (x_0)$ $x_0 = 1.47 \text{ mm}$
9(b)(ii)	The amplitude would be the same. The mass of the pebbles would not affect the acceleration and minimum amplitude.
9(c)(i)1.	$E_T = KE_{\max}$ $= \frac{1}{2} m \omega^2 x_0^2$ $= \frac{1}{2} (1.2) (2\pi f)^2 (0.034)^2$ $= 0.171 \text{ J}$
9(c)(i)2.	$KE = PE$ $\frac{1}{2} m \omega^2 (x_0^2 - x^2) = \frac{1}{2} m \omega^2 x^2$ $x = \frac{x_0}{\sqrt{2}} = 0.024 \text{ m} = 2.4 \text{ cm}$
9(c)(ii)	